

Akshay Kaushik

Machine Learning Engineer — LLM, RAG & Agent Systems

Punjab, India | +91-9876635806 | 2021pcs1016@iitjammu.ac.in
aktyzon.github.io/Akshay-portfolio | github.com/AkTyzon | linkedin.com/in/akshay-kaushik

Three years building retrieval and agent systems for security operations — natural-language querying over live production data, anomaly detection across millions of daily events, and the evaluation tooling to prove any of it works. Six systems deployed and publicly reachable.

EXPERIENCE

Centre for Development of Telematics (C-DOT)

Aug 2023 – Present · New Delhi

Scientist B — Machine Learning / AI Engineer

Natural-Language Security Analytics Platform

- Architected a multi-turn **ReAct agent** translating analyst questions into grounded PostgreSQL and OpenSearch queries, orchestrated as a **plan → execute → synthesize → review** state machine with selected capabilities executing **concurrently**, so multi-source questions cost the slower path rather than the sum. Capabilities self-describe their routing rules, availability and data scope, and the planner prompt is generated from that registry — so the system cannot advertise a capability it is unable to run.
- Deploys against a different organization's data with no code change** — no table, column or rule name exists in source; the entire surface (**40 PostgreSQL tables, ~1,700 described OpenSearch fields**) resolves at runtime from per-deployment catalogs, enforced by an automated audit.
- Designed the correctness layer — review checkpoint for answers built on unresolved intermediate results, deterministic pre-execution query validation, scope-aware judge blocking unsupported claims. Cut worst-case turnaround **232s → 66s**; wrote **62 automated tests from zero** plus a regression benchmark diffed before and after every change.

Large-Scale Anomaly Detection

- Reduced false-positive security alerts **30%** via feature engineering and threshold calibration across **5M+ events per endpoint per day**; shipped containerized inference with automated retraining, plus classifiers detecting processes impersonating legitimate system binaries (allowlists + Levenshtein distance).

Enterprise Knowledge Assistant

- LLM-powered question answering over **1,000+ internal policy, technical and operational documents**, with source citations and downloadable references attached to every answer so staff can verify rather than trust.

PROJECTS

EvalForge — LLM Evaluation & Benchmarking Platform

FastAPI · PostgreSQL · Qdrant · LangChain · RAGAS · DeepEval · Docker · GCP

[Live demo](#) · [Live API](#)

- Scores every answer on **18 metrics** — classical NLP (BLEU, ROUGE, METEOR), retrieval quality (Recall@K, MRR, NDCG) and LLM-as-judge faithfulness — with statement-level analysis isolating which claims the retrieved context does not support.
- Provisioned and operate the full deployment (containerized services, managed relational and vector stores, auto-renewing TLS); its own regression harness caught a token-budgeting change that degraded quality, reverted before it shipped.

Agentic Research Assistant

FastAPI · sentence-transformers · Groq (Llama 3.3 70B) · arXiv API

[Live demo](#)

- RAG over arXiv with inline citations, which declines to answer when the retrieved passages do not support one rather than filling the gap.
- Diagnosed two retrieval defects behind fluent but ungrounded output: unquoted arXiv phrase search returned off-topic papers, and embedding the question without its topic left cosine similarity at **0.08** — retrieving on topic + question raised it to **0.40**, no model change.

Additional Projects

- Cascade Learning on Spark** (PySpark, Spark MLlib, 581K rows) — a logistic stage answers what it is confident about and only the residual escalates to a random forest: retains **99.1%** of the forest's accuracy while avoiding **44%** of its inference cost, characterised across nine confidence thresholds. [Results](#)
- Recommender systems** (scikit-learn, MovieLens) — item-item collaborative filtering, leave-last-out: **15.5% hit-rate@10 vs a 5.9% popularity baseline** ($2.63\times$ lift); content-based recipe similarity at **96% tag agreement@5 vs 38.5% random**. [Movie](#) · [Recipe](#)

SKILLS

ML & AI	LLMs, RAG, agentic systems (ReAct), prompt engineering, anomaly detection, recommender systems, deep learning
Languages	Python, C++, SQL
LLM / NLP	PyTorch, LangChain, LangGraph, HuggingFace, sentence-transformers, Qdrant, FAISS, RAGAS, DeepEval
Data & Infra	PostgreSQL, OpenSearch, PySpark, Spark MLlib, Docker, FastAPI, Linux, Git, GCP (Compute Engine, self-hosted deployments, automated TLS)

EDUCATION, AWARDS & CERTIFICATIONS

M.Tech, Data Science — Indian Institute of Technology (IIT) Jammu CGPA 8.48 · 2023
B.Tech, Computer Science — Chandigarh University CGPA 7.8 · 2020

- PromptWars** (Google for Developers · Build with AI) — top 400 of 2,565 participants, 2026
- IBM** — Develop Generative AI Applications · **IBM** — Agentic AI with LangGraph, CrewAI, AutoGen & BeeAI